The Astrophysical Journal Letters, 811:L7 (5pp), 2015 September 20 doi:10.1088/2041-8205/811/1/L7
© 2015. The American Astronomical Society. All rights reserved.

STRAHL FORMATION IN THE SOLAR WIND ELECTRONS VIA WHISTLER INSTABILITY

Jungjoon Seough[1,5], Yasuhiro Nariyuki[1], Peter H. Yoon[2,3], and Shinji Saito[4]
[1] Faculty of Human Development, University of Toyama, 3190, Gofuku, Toyama City, Toyama 930-8555, Japan
[2] Institute for Physical Science and Technology, University of Maryland, College Park, MD 20742, USA
[3] School of Space Research, Kyung Hee University, Yongin-Si, Gyeonggi-Do 446-701, Korea
[4] Graduate School of Science, Nagoya University, Furocho, Chikusa, Nagoya 464-8601, Japan
Received 2015 June 15; accepted 2015 August 8; published 2015 September 16

ABSTRACT

This Letter puts forth a possible explanation for the formation of a solar wind strahl electron distribution function by means of local wave–particle interactions. A solar wind electron consists of a core and the hot "halo" electrons, which possess a net drift speed with respect to the core. According to the present model, pitch angle scattering of the initially isotropic drifting halo occurs when the enhanced whistler waves are excited by mildly anisotropic core electrons. The pitch angle scattering primarily affects the halo moving in the anti-sunward direction, resulting in pitch angle diffusion across the 90° gap through a nonlinear scattering process, and consequently leading to a reduction in the net drift speed of halo electrons. The remaining portion of the anti-sunward moving halo, which is not affected by pitch angle scattering, simply appears to form a field-aligned strahl in the electron velocity distribution. The present scenario of local generation of the strahl is demonstrated by the particle-in-cell simulation.

Key words: instabilities – interplanetary medium – plasmas – solar wind – waves

1. INTRODUCTION

Electrons in the interplanetary space exhibit significant nonthermal features such as the temperature anisotropy, heat flux, and different populations including the thermal core and two distinct suprathermal components: the halo and the strahl (Feldman et al. 1975; Pilipp et al. 1987; Marsch 2006; Štverak et al. 2008). Coulomb collisions are important for regulating the overall electron properties, but those effects on the suprathermal components are relatively insignificant due to sufficiently long mean free paths (Salem et al. 2003). Electrons play large roles in driving the expansion of solar corona and transporting solar wind energy through the heat flux (Feldman et al. 1975; Marsch 2006). They may provide important clues about the solar wind acceleration or coronal plasma conditions.

Suprathermal electrons are discerned through a significant break or a sudden change in the slope of the measured electron velocity distribution functions (eVDFs), rendering them distinguishable from the core with lower kinetic energy at a few tens of electron volts. The halo and strahl have very different properties in velocity space. The halo is present at all pitch angles, whereas the strahl, the most striking feature in the eVDF, appears as a magnetic-field-aligned beam streaming away from the Sun. It is believed that the halo originates from the solar corona where hot electrons are continuously escaping from the Sun at speeds higher than those of the core electrons. Even though the strahl is typically associated with the high-speed solar wind, its origin, either in the solar corona or interplanetary space, has not yet been fully understood.

Several theoretical models have been suggested to explain how the strahl is generated (Scudder & Olbert 1979; Lie-Svendsen et al. 1997; Pierrad et al. 1999; Landi et al. 2012; Smith et al. 2012). Basically, they focus on the role of Coulomb collisions and expanding magnetic field configurations. Vocks & Mann (2003), on the other hand, suggest that the combined effects of magnetic mirror force and the whistler wave-induced diffusion may play a role in generating suprathermal electrons that includes the strahl (Vocks & Mann 2003). However, the numerically computed strahl shows an extremely narrow feature, which is not consistent with the observations (Hammond et al. 1996; Anderson et al. 2012), and which necessarily implies that certain processes in interplanetary space must be present in order to efficiently scatter and broaden the strahl component (Vocks et al. 2005; Saito & Gary 2007; Pavan et al. 2013).

In situ measurements covering the heliocentric distance from 0.3 up to 4 AU have shown the opposite trend between the halo and strahl relative densities (Maksimovic et al. 2005; Štverak et al. 2009). The relative number of strahl decreases with heliocentric distance from the Sun, while that of the halo increases, suggesting that the halo consists partly of electron pitch angle scattered out of the strahl through presumed plasma waves, but the precise nature of the plasma waves responsible for the scattering remains unidentified.

In this Letter, we put forth an alternative explanation for the origin and formation of the strahl component. We begin by noting two observational facts that recently became available through statistical analyses of the *Wind* spacecraft data at 1 AU. First, in the collisionless plasmas, the observed value of the electron core–proton drift is large (up to 200 km s^{-1}; Pulupa et al. 2014). This can be understood as the less-abundant halo electrons flowing much faster than the protons or core electrons in order to maintain the zero net current. Such a parallel drift has been found typically in high-speed solar wind, which is a favorable condition for the observed core–halo–strahl feature. Second, the thermal core is not so isotropic but its temperature anisotropies are bounded by the theoretical thresholds of the whistler (for $T_{\perp c} > T_{\parallel c}$) and the fire-hose instabilities (for $T_{\perp c} < T_{\parallel c}$), where $T_{\perp c}$ and $T_{\parallel c}$ are perpendicular and parallel temperatures of the core electrons defined with respect to the ambient magnetic field (Gary & Wang 1996; Gary & Nishimura 2003; Štverak et al. 2008; C. Salem 2015, private

[5] International Research Fellow of the Japan Society for the Promotion of Science.

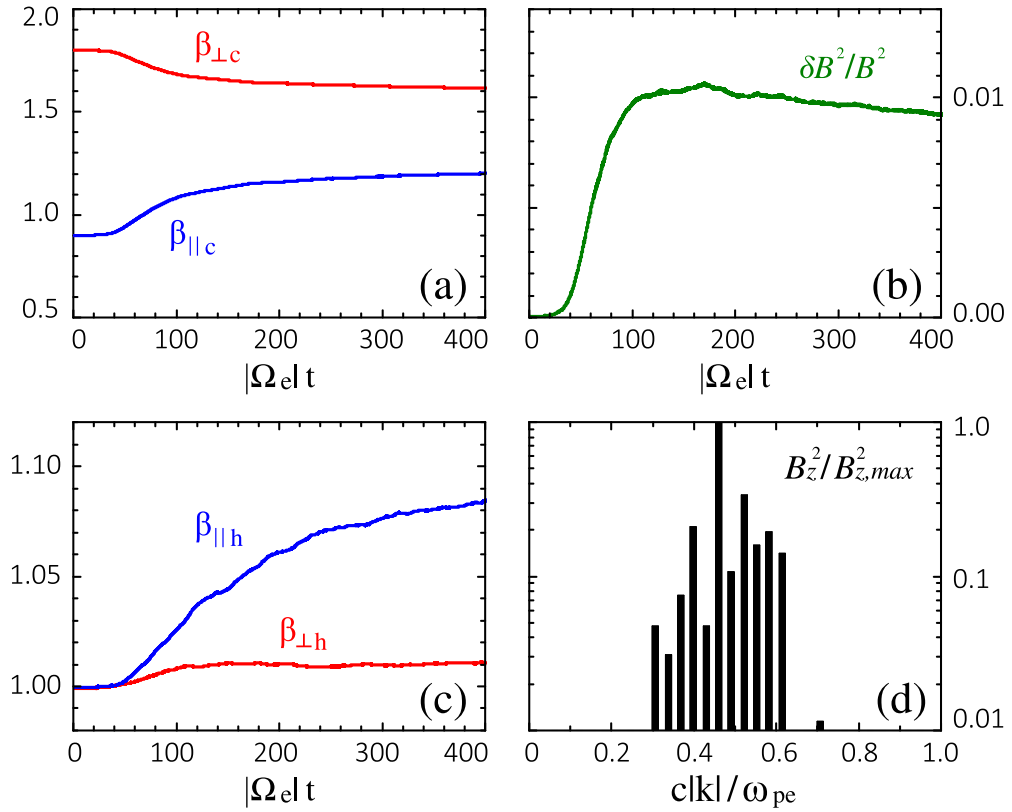


Figure 1. Time evolution of the simulated system: (a) normalized perpendicular and parallel temperatures of the core, (b) normalized fluctuating magnetic field energy density, (c) normalized perpendicular and parallel temperatures of the halo, and (d) the power spectrum of the z-component of the fluctuating magnetic field at $|\Omega_e|t = 200$.

communication). This provides strong evidence that these instabilities are operative in solar wind electrons.

If instabilities driven by core temperature anisotropy occur, then the resulting enhanced fluctuations could lead to pitch angle scattering of the drifting halo in an asymmetric way; this is because the number of halo particles participating in the wave–particle interaction is different between the halo moving in the sunward and anti-sunward directions in the proton rest frame. Such a differential pitch angle scattering mechanism may provide a possible explanation for the origin and formation process of the strahl. In the remainder of this Letter, we demonstrate this process by a particle-in-cell simulation method.

2. STRAHL FORMULATION AND NUMERICAL ANALYSIS

2.1. Numerical Setup

We set a one-dimensional particle-in-cell simulation in the solar wind proton frame of reference with a uniform magnetic field $\boldsymbol{B} = B\hat{x}$, which is directed away from the Sun. An electron–proton plasma is uniformly distributed in space. Initially, the proton VDF is a single isotropic Maxwellian. The electron VDF is a combination of two drifting components in the proton rest frame; the VDFs for core and halo are anisotropic bi-Maxwellian and isotropic Maxwellian, respectively. Thus, the overall electron VDF is initially given by

$$f_e(v_\parallel, v_\perp) = \left(\frac{m_e}{2\pi}\right)^{3/2}\left[\frac{1-\delta}{T_{\perp c}T_{\parallel c}^{1/2}}\exp\left(-\frac{m_e v_{\perp c}^2}{2T_{\perp c}} - \frac{m_e\left(v_{\parallel c}-V_{cd}\right)^2}{2T_{\parallel c}}\right) + \frac{\delta}{T_{\perp h}T_{\parallel h}^{1/2}}\times\exp\left(-\frac{m_e v_{\perp h}^2}{2T_{\perp h}} - \frac{m_e\left(v_{\parallel h}-V_{hd}\right)^2}{2T_{\parallel h}}\right)\right], \quad (1)$$

where $v_{\perp a}$ and $v_{\parallel a}$ are thermal velocity components perpendicular and parallel to the ambient magnetic field, respectively, and m_e is the electron mass. $T_{\perp a}$ and $T_{\perp a}$ are perpendicular and parallel temperatures. Here, the subscript label a = e, c, h, p stands for overall electron, core, halo, and proton. $\delta = n_h/n_e$ is the relative halo density normalized by the total electron density, $n_e = n_c + n_h$, and V_{cd} and V_{hd} are relative drift speeds of the core and halo, respectively. These speeds satisfy $\delta V_{hd} = -(1-\delta)V_{cd}$, which guarantees the zero net current. Guided by observations (Maksimovic et al. 2000; Pulupa et al. 2014), we choose $\delta = 0.1$ and $V_{hd} = 22.5v_A$ so that $V_{cd} = -2.5v_A$, where v_A is the Alfvén speed. For the whistler instability driven by the core temperature anisotropy, we choose the following physical parameters: anisotropic core

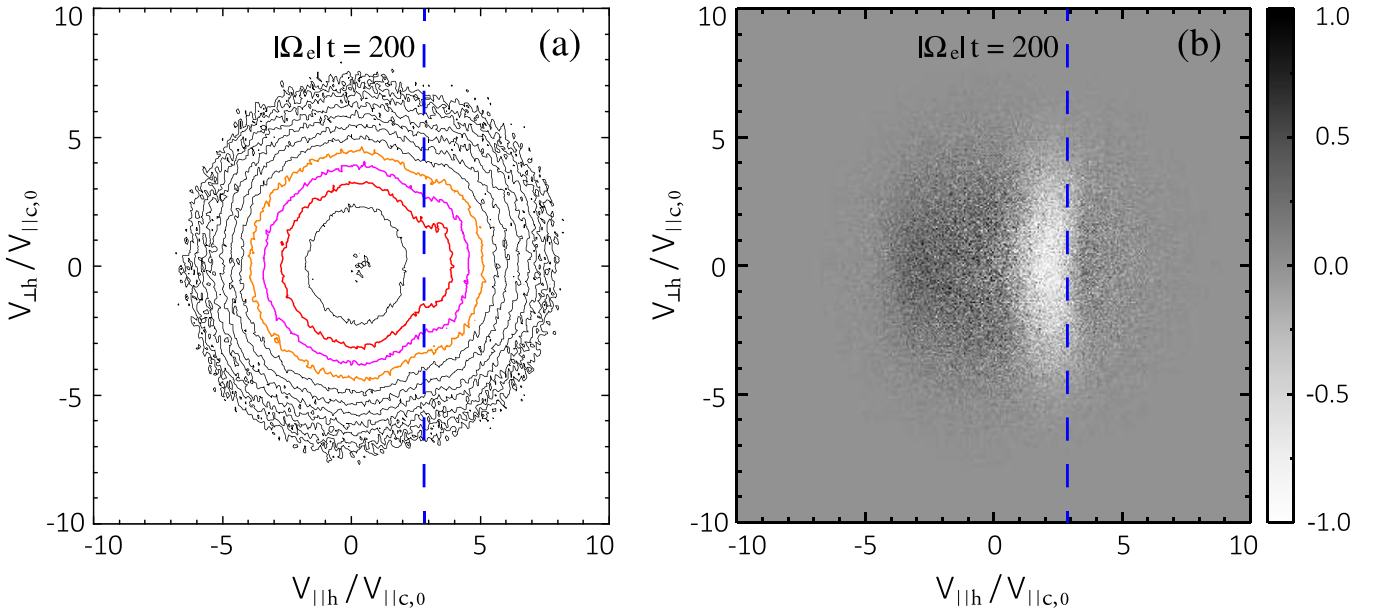


Figure 2. (a) Halo VDF at $|\Omega_e|t = 200$ in the $(v_\parallel, v_\perp)$ plane. (b) Temporal difference between the halo VDFs at different times, $\Delta f_h(v_\parallel, v_\perp) = f_h(v_\parallel, v_\perp)|_{\Omega_e|t=200} - f_h(v_\parallel, v_\perp)|_{\Omega_e|t=0}$.

$T_{\perp c}/T_{\parallel c} = 2$, core parallel plasma beta $\beta_{\parallel c} = 0.9$, isotropic halo $T_{\perp h}/T_{\parallel h} = 1$, halo parallel beta $\beta_{\parallel h} = 1$, halo-to-core temperature ratio $T_{\parallel h}/T_{\parallel c} = 10$, and isotropic proton temperature $T_p/T_{\parallel c} = 1$, respectively, where the parallel beta is defined by $\beta_{\parallel a} = 8\pi n_a T_{\parallel a}/B^2$. The simulation system contains 4096 grid points with the grid spacing $\Delta x = 0.05c/\omega_{pe}$. The time step is $\Delta t = 0.002/|\Omega_e|$, $|\Omega_e|/\omega_e = 0.08$, the proton-to-electron mass ratio is $m_p/m_e = 1600$, and the number of particles per cell for each species is 16,000. Here, the electron plasma frequency is $\omega_{pe} = \sqrt{4\pi n_e e^2/m_e}$, and the electron cyclotron frequency is $\Omega_e = -eB/m_e c$, with e, B, and c denoting the unit electric charge, the magnetic field intensity, and the speed of light in vacuo, respectively. The present initial condition with the relative high core–halo drift is subject to the whistler heat flux instability (Gary et al. 1994), but the growth rate is much lower than that of the whistler anisotropy instability.

2.2. Numerical Results

Figure 1 shows the time evolution of the core and halo plasma betas, the fluctuating magnetic field energy density, $\delta B^2/B^2$, and the fluctuation spectrum at $|\Omega_e|t = 200$. Figures 1(a) and (b) are self-explanatory, and they show the well-known dynamical properties associated with the whistler instability. The core anisotropy and the fluctuating magnetic field energy density show relatively little change after the quasilinear saturation. The temporal evolutions of both the perpendicular and parallel temperatures of an initially isotropic drifting halo are shown in Figure 1(c). The increase of perpendicular halo temperature goes on until the saturation of the whistler instability occurs at $|\Omega_e|t \approx 120$, and then it stops and eventually stays nearly constant. This perpendicular heating is due to the resonant interactions between halo particles and whistler waves, resulting in the pitch angle diffusion in phase space. It is interesting to note, however, that the parallel halo temperature continuously increases over time.

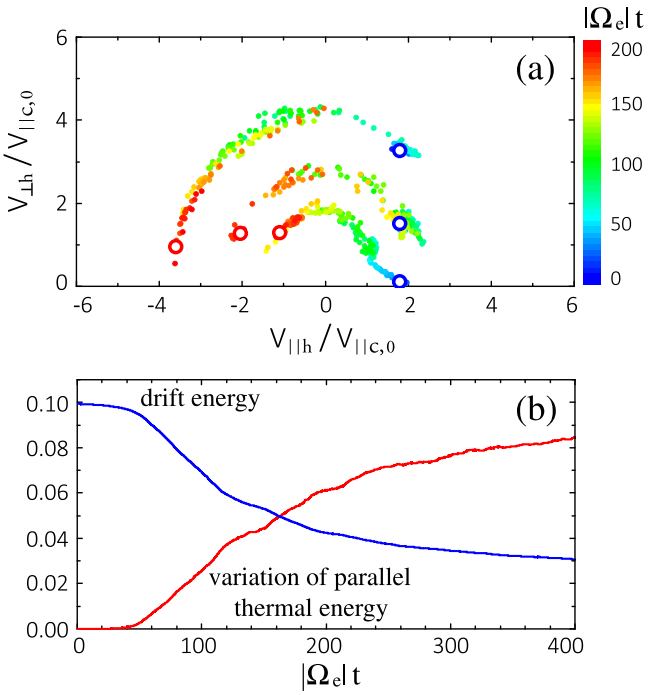


Figure 3. (a) Sample trajectories of the halo particles that manage to diffuse across the 90° pitch angle barrier in velocity space. (b) The time evolution of the halo drift energy (blue) and the variation of the parallel thermal energy (red) of the halo. Both energies are normalized to the initial parallel thermal energy of the halo.

This behavior may be related to an intrinsically nonlinear effect because the parallel heating of the halo continues further even after the quasilinear saturation of the waves.

In order to investigate how the enhanced whistler waves scatter the drifting halo particles, we focus on the halo VDF, $f_h(v_\parallel, v_\perp)$. The scattered $f_h(v_\parallel, v_\perp)$ at $|\Omega_e|t = 200$, shown in Figure 2(a), deviates far from the initial shifted-Maxwellian

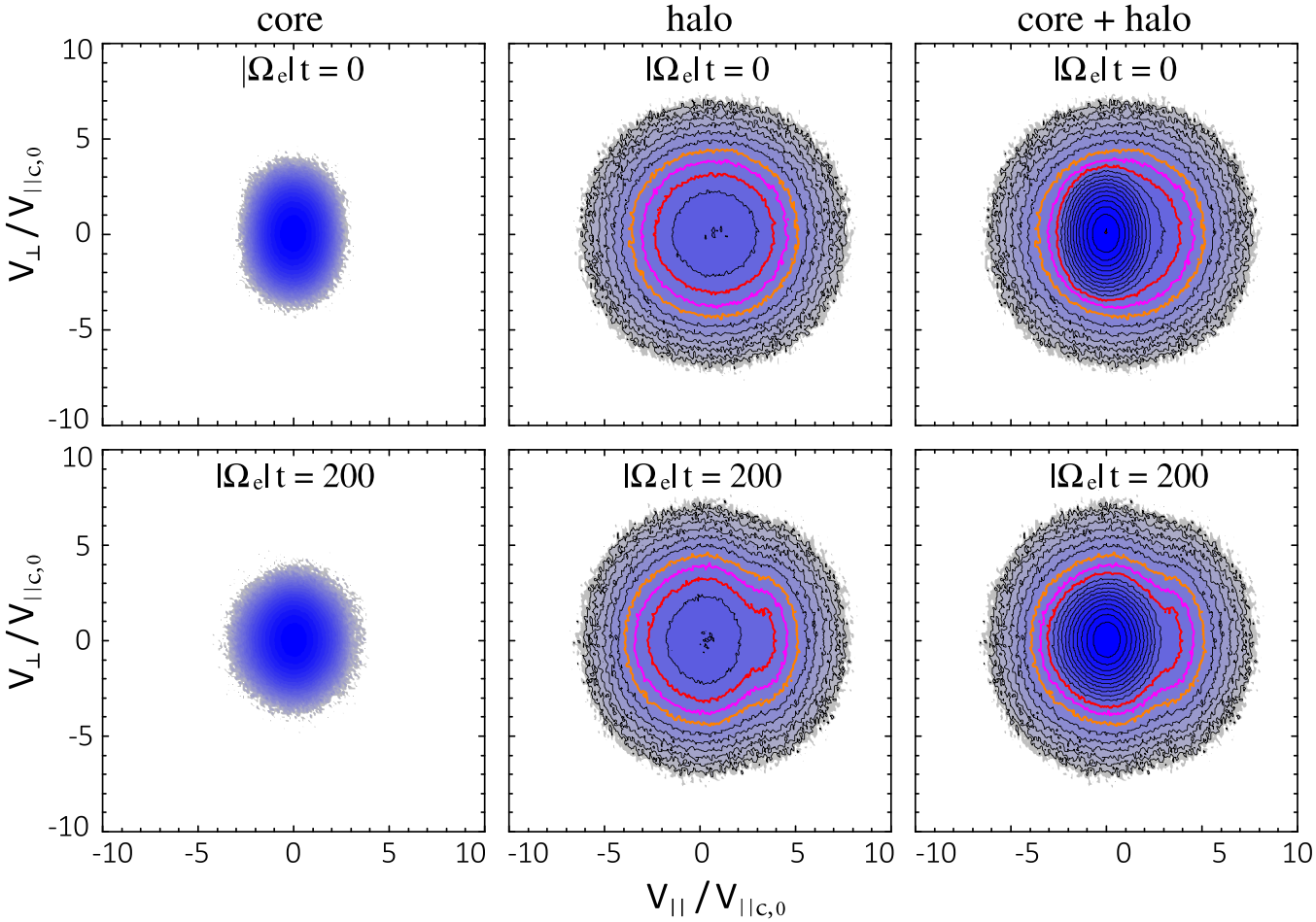


Figure 4. Top: contour plots of the core, halo, and overall electron VDFs in ($v_\parallel$, $v_\perp$) plane at the initial state. Bottom: contour plots of the core, halo, and total electron VDFs in ($v_\parallel$, $v_\perp$) plane at $|\Omega_e|t = 200$.

form and has a highly asymmetric configuration in parallel velocity, where the horizontal and vertical axes are the halo velocity components normalized to the initial core parallel thermal velocity $v_{\parallel c,0}$. The most striking feature of the halo VDF is that an apparently beam-like component superficially resembling the broadened strahl is seen to be streaming away from the Sun along the magnetic field direction at $v_\parallel > 0$ (highlighted colors). Note that such a feature is a result of local wave–particle interaction and that the apparent halo and strahl are actually one and the same, except that the strahl feature is formed as a result of pitch angle scattering of the halo. The vertical blue dashed line, for reference, is the maximum value of the parallel resonance velocity $v_{\parallel h}/v_{\parallel c} \approx 2.86$ that corresponds to the cyclotron resonance condition $v_\parallel = (\omega - |\Omega_e|)/k_\parallel$ within a broad spectrum of the enhanced fluctuations at $|\Omega_e|t = 200$, as shown in Figure 1(d); here, ω and $k_\parallel$ are the wave angular frequency and parallel wavenumber, respectively. The corresponding normalized values of the resonance frequency and wavenumber are $\omega/|\Omega_e| \approx 0.123$ and $ck_\parallel/\omega_{pe} \approx -0.3$. The halo VDF in the range $|v_{\parallel h}|/v_{\parallel c} \lesssim 2.86$ appears to possess a slight perpendicular anisotropy resulting from the resonant interaction, which is responsible for an increase of halo pitch angle, often to 90°. It is obvious, however, that the linear cyclotron resonance process in the quasilinear diffusion model is insufficient to account for the resulting field-aligned strahl.

Figure 2(b) shows the temporal difference between the halo VDFs at different times, $\Delta f_h(v_\parallel, v_\perp) = f_h(v_\parallel, v_\perp)|_{|\Omega_e|t=200} - f_h(v_\parallel, v_\perp)|_{|\Omega_e|t=0}$. Positive and negative values indicate an increase and decrease in the number of halo particles in velocity space. As one can see, the halo responds asymmetrically with respect to $v_\parallel = 0$. The sign of Δf_h is opposite for $v_\parallel > 0$ and $v_\parallel < 0$. This is mainly because the halo particles at $v_\parallel > 0$ are forced to diffuse across the $v_\parallel = 0$ via non-resonant pitch angle scattering, namely, the Lorentz force by nonlinear whistler waves. Note that the region corresponding to the negative values in Figure 2(b) mostly overlaps with the resonant particles. In the standard quasilinear theory, pitch angle diffusion across 90° is strictly prohibited. The Lorentz force that is expressed by the cross product of the wave magnetic field and the perpendicular velocity acts on some resonant halo particles drifting anti-sunward direction to cross the $v_\parallel = 0$ barrier. The remaining halo electrons that fail to cross the $v_\parallel = 0$ barrier appear as the apparent strahl in the halo VDF.

Figure 3(a) shows sample trajectories of the halo particles that manage to diffuse across the 90° pitch angle gap in velocity space. The trajectories in Figure 3(a) show the time evolution of the initial states corresponding to $[v_\parallel(0), v_\perp(0)] = (1.8, 0.1)$, $(1.8, 1.5)$, and $(1.8, 3.26)$, respectively. The increase in time is depicted by the color map scale. The open circles denote the halo locations at the initial (blue)

and later times (red), $|\Omega_e|t = 200$. Both resonant and non-resonant scattering processes occur in velocity space. Note, however, that the resultant diffusion processes cannot adequately be explained by the quasilinear cyclotron resonance alone. The individual trajectories obviously show that the Lorentz force exerted on the drifting halo particles leads to changing the sign of the velocity along $v_\parallel$ as they move back and forth along the diffusion paths. Such particle diffusion processes by nonlinear counter-propagating waves are stronger than those predicted by the quasilinear theory (Kuramitsu & Hada 2000).

Since a portion of drifting halo can diffuse through the 90° pitch angle, the halo drift energy is expected to be reduced. This is evidenced by Figure 3(b), which shows the time history of the halo drift energy (blue) and the variation of the parallel thermal energy (red) of the halo. Here, both energies are normalized to the initial parallel thermal energy of the halo. The temporal evolution of these energies clearly shows an opposite trend; the drift energy decreases over time after the whistler waves begin to grow, while the parallel thermal energy increases at the expense of the drift energy. This energy exchange occurs simply because of the Lorentz force exerted on the halo particles. Such a decrease in the drift energy is involved in the reduction in a net drift speed of the halo, resulting in a shift of the peak of the halo VDF toward the sunward direction with the drift speed, one-half of its initial value. Here, we point out that the energy transfer is an intrinsically nonlinear phenomenon, which is responsible for the parallel halo heating, as shown in Figure 1(c).

Figure 4 displays contour plots of the core, halo, and total electron VDFs in the $(v_\parallel, v_\perp)$ plane at $|\Omega_e|t = 0$ (top panels) and 200 (bottom panels). The core VDF retains the bi-Maxwellian shape, albeit with reduced anisotropy, while the halo VDF shows a distinct departure from its initial Maxwellian and yields the strahl formation. The net electron VDF shown in the bottom right panel of Figure 4 is remarkably similar to the core–halo–strahl feature observed in the solar wind electrons as shown in the top panel of Figure 1 of Marsch (2006).

3. SUMMARY AND DISCUSSION

In summary, we have presented a particle-in-cell simulation of the whistler instability driven by the anisotropic core electron in the presence of the drifting halo. Combined effects of the core–halo with relatively large drift and the whistler instability are demonstrated to generate a field-aligned strahl through non-resonant pitch angle scattering of an anti-sunward moving halo. Nonlinear pitch angle scattering randomizes the halo orbits, enabling halo particles to diffuse over the 90° pitch angle gap in velocity space, accompanied by the energy transfer from the drift energy to the parallel thermal energy of the drifting halo. The concomitant reduction in a net drift speed of the halo plays a role in distinguishing the strahl feature from the halo electrons in velocity space. As a result, the remaining portion of the halo particles, which are not affected by pitch angle scattering, forms the strahl. The resultant core–halo–strahl feature in our simulation is consistent with the observed VDF in the solar wind electron as presented in Figure 1 of Pilipp et al. (1987).

Before we close, we point out that the measured energy of the field-aligned strahl often extends up to 1 keV near 1 AU. However, the present simulation model does not address such an energetic strahl component since the simulated eVDF exhibits a strahl-like feature at approximately two to five times the core thermal speed corresponding to the kinetic energies of a few hundred eV. We also do not take into account the importance of global-scale dynamics such as collisional effects and expanding magnetic field configuration, which have been proposed by several theoretical models. Nevertheless, although the measured high-energy strahl component cannot be accounted for in the present simulation, we have shown that the scenario of the local generation of the strahl is one of the possible mechanisms that may contribute to the strahl formation in the solar wind electron distribution functions.

J.S. and Y.N. were supported by the Postdoctoral Fellowship for Foreign Researchers of the Japan Society for the Promotion of Science (grant No. P14028). J.S. thanks Y. Omura for providing KEMPO code used in this work and useful discussion and also thanks C. Salem for fruitful discussions. P.H.Y. acknowledges NSF grants AGS1138720 and AGS1242331 to the University of Maryland, and the BK21-Plus grant to Kyung Hee University, Korea, from the National Research Foundation (NRF) funded by the Ministry of Education of Korea.

REFERENCES

Anderson, B. R., Skoug, R. M., Steinberg, J. T., & McComas, D. J. 2012, JGR, 117, A04107
Feldman, W. C., Asbridge, J. R., Bame, S. J., Montgomery, M. D., & Gary, S. P. 1975, JGR, 80, 4181
Gary, S. P., & Nishimura, K. 2003, PhPl, 10, 3571
Gary, S. P., Scime, E. E., Phillps, J. P., & Feldman, W. C. 1994, JGR, 99, 23391
Gary, S. P., & Wang, J. 1996, JGR, 101, 10749
Hammond, C. M., Feldman, W. C., McComas, D. J., Phillips, J. L., & Foryth, R. J. 1996, A&A, 316, 350
Kuramitsu, Y., & Hada, T. 2000, GeoRL, 27, 629
Landi, S., Matteini, L., & Pantellini, F. 2012, ApJ, 760, 143
Lie-Svendsen, Ø., Hansteen, V. H., & Leer, E. 1997, JGR, 102, 4701
Maksimovic, M., Gary, S. P., & Skoug, R. M. 2000, JGR, 105, 18337
Maksimovic, M., Zouganelis, I., Chaufray, J.-Y., et al. 2005, JGR, 110, A09104
Marsch, E. 2006, LRSP, 3, 1
Pavan, J., Vinas, A. F., Yoon, P. H., Ziebell, L. F., & Gaelzer, L. 2013, ApJL, 769, L30
Pierrad, V., Maksimovic, M., & Lemaire, J. 1999, JGR, 104, 17021
Pilipp, W. G., Miggenrieder, H., Mühlhäuser, K.-H., et al. 1987, JGR, 92, 1075
Pulupa, M., Bale, S., Salem, C., & Horaites, K. 2014, JGR, 119, 647
Saito, S., & Gary, S. P. 2007, GeoRL, 23, L01102
Salem, C., Hubert, D., Lacombe, C., et al. 2003, ApJ, 585, 1147
Scudder, J. D., & Olbert, S. 1979, JGR, 84, 2755
Smith, H. M., Marsch, E., & Helander, P. 2012, ApJ, 753, 31
Štverak, Š., Maksimovic, M., Travnicek, P. M., et al. 2009, JGR, 114, A05104
Štverak, Š., Trávníček, P. M., Maksimovic, M., et al. 2008, JGR, 113, A03103
Vocks, C., & Mann, G. 2003, ApJ, 593, 1134
Vocks, C., Salem, C., Lin, R. P., & Mann, G. 2005, ApJ, 627, 540